Ram Calculus and Brennan's Conjecture

M150: Exact Integral-Means Coordinates, Critical-Tilt Obstructions, and Certified Transfer and Transport Bounds

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M150

October 7, 2026

Abstract

OpenAI's September 24 manuscript on Brennan's conjecture claims the sharp inverse-square integral-means exponent for the full normalized univalent class. We analyze its transfer operator, weighted affine law, escape estimates, critical-point pairing, and transport contradiction, and identify concrete interfaces with the Ram log-partition, stable-frame, and physical-coordinate methods. The exact log-partition identity survives this comparison, but an earlier proposed pointwise tilted-variance premise does not: for the bounded univalent quadratic $f(z) = z - z^2/2$, variance at tilt -1 grows as $T^2/12 + O(T)$, where $T = \log(1/(1-r))$. We prove this obstruction and replace the premise by an averaged criterion that allows curvature concentration. We give reciprocal-derivative coefficient certificates, positive transfer superfunctions with residual margins, an explicit spectral lower calibration, and a quantitative finite-box version of the source's transport comparison. A finite law matching the displayed root moments but violating affine covariance demonstrates why moment fitting alone cannot reproduce the proof. These are rigorous local results and conditional certification tools, not a second independent proof of the full Brennan assertion. The report separates source claims, verified calculations, external theorem dependencies, and remaining all-class and all-scale obligations.

Contents

1 Problem, Classes, and Evidence Standard **3**
1.1 Exact target . . . 3
1.2 Source and audit boundary . . . 3
1.3 Current public comparisons . . . 3

2 Reconstructing the Source's Main Mechanism **4**
2.1 The genuine state space and transfer operator . . . 4
2.2 A law under the contrary growth hypothesis . . . 4
2.3 Escape, topology, and transport . . . 5
2.4 Area range and endpoint limits . . . 6

3 Exact Ram Log-Partition Coordinates **6**

4 A Critical-Tilt Obstruction to an Earlier Premise **7**

5 Transfer Certificates and a Moment-Matching Obstruction **8**
5.1 Positive certificates in the correct norm 8
5.2 Why root moments cannot replace covariance 8

6 Certified Ram Panels **9**
6.1 Reciprocal-derivative coefficients 9
6.2 Log-field panels and asymptotic budgets 9
6.3 A separate Schwarzian-multiplier route 10

7 A Quantitative Transport Extension **10**

8 Further Deductions and Limits **11**
8.1 Loss-free endpoint bounds require another theorem 11
8.2 Negative tails and bounded versus full classes 11

9 Comparison, Research Directions, and Conclusion **11**
9.1 Advantages and costs 11
9.2 Focused future work 12
9.3 Conclusion 12

A Reproducibility and Source Locators **12**

1 Problem, Classes, and Evidence Standard

1.1 Exact target

Let $\mathcal{S}$ consist of injective holomorphic $f : \mathbb{D} \to \mathbb{C}$ normalized by $f(0) = 0$ and $f'(0) = 1$. Let $\mathcal{S}_b$ be its bounded-image subclass. For real p, put

$$I_f(p,r) = \frac{1}{2\pi}\int_{-\pi}^{\pi} |f'(re^{i\theta})|^p \,\mathrm{d}\theta, \quad T = \log\frac{1}{1-r}, \quad \beta_f(p) = \limsup_{r\uparrow 1} \frac{\log I_f(p,r)}{T}. \tag{1}$$

Define $B_{\mathcal{S}}(p) = \sup_{f\in\mathcal{S}} \beta_f(p)$ and B_b similarly. The source M150 [1] asserts

$$\forall \epsilon > 0 \; \exists C_\epsilon < \infty \; \forall f \in \mathcal{S} \; \forall r \in [1/2, 1) : \quad I_f(-2,r) \le C_\epsilon (1-r)^{-1-\epsilon}, \qquad B_{\mathcal{S}}(-2) = 1. \tag{2}$$

No smooth boundary, bounded image, or quasiconformal extension is assumed. For a conformal bijection $\varphi : W \to \mathbb{D}$, with inverse f, nonnegative change of variables gives

$$\int_W |\varphi'|^s \,\mathrm{d}A = \int_{\mathbb{D}} |f'|^{2-s} \,\mathrm{d}A. \tag{3}$$

Thus Brennan's proposed open interval $4/3 < s < 4$ corresponds to $-2 < t < 2/3$, $t = 2 - s$. Here W is a simply connected plane domain with at least two boundary points on the sphere. Endpoint inclusion is false in general; neither an exponent of one nor an arbitrary epsilon loss implies area integrability at the inverse-square endpoint.

The order of quantifiers matters. A uniform bound such as (2) implies the upper spectrum bound, but a diagonal optimizer changing f with r is not a fixed-map lower spectrum certificate. A finite list of maps does not establish an upper bound on the full class.

1.2 Source and audit boundary

M150 is family 72 in the collection pinned to commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, acquired October 7, 2026. An independent task-local audit read every proof in its six technical sections, the introduction/history, and supporting source files. The parent analysis reread the statement, weighted-law construction, transport proof, and area deductions and derived the local results below. No demonstrable fatal gap was identified in the inspected main chain. This is not formal verification, a referee decision, or a guarantee of the claimed theorem. In particular, not finding a counterexample is not a proof of correctness.

The main local Ram source is TR-item-077-astra [9]. Its log-partition identities are directly applicable, while its bounded-class spectrum and conditional closure proposals require the distinctions below. TR248 [10] contributes finite reconstruction and leakage accounting; TR250 [11] contributes physical norms under moment-coordinate changes. These tools do not supply univalence, affine covariance, or sharp global growth merely by renaming coordinates. Only the particular corpus results actually used are cited; no claim of auditing every private report is made.

1.3 Current public comparisons

The relevant public literature has several different scopes. Jin's June 2026 version [2] states Brennan for univalent rational maps and recalls $B_{\mathcal{S}}(p) = \max\{B_b(p), 3p-1\}$ for real p; its negative spectra consequently agree. Aleman–Kouroupis [4] establishes the derivative-integrability interval for conformal semigroups and their Koenigs maps. Zheng [5] treats specified simply connected polynomial basins, including a small-coefficient hypothesis for one family. These are restricted results, not substitutes for M150's universal claim. Their relevant statements, rather than all their proofs, were checked.

Classical weighted-Bergman estimates include the value 1.218 at $p = -2$ in Hedenmalm–Shimorin [6]; Sola's thesis [7] reports 1.206. These are historical comparisons, not a claim that a complete October 2026 best-bound search has been established. Shimorin's multiplier criterion, recalled by Jin, [3] is an additional sufficient route, discussed in Section 6. A focused primary-source search found no independently audited replacement proof that would remove the need to scrutinize the new release. It does not establish exhaustive literature coverage or community acceptance.

2 Reconstructing the Source's Main Mechanism

2.1 The genuine state space and transfer operator

Write $m(z) = (z - i)/(z + i)$ and $F(z) = 2if(m(z))$. Then $F(i) = 0$, $F'(i) = 1$. The normalized half-plane univalent class $\mathcal{K}$ is compact in local uniform convergence. Normalization prevents constant limits, and Hurwitz preserves injectivity. Set $Q_F = 1/F'$. For $z = x + iy$, let

$$T_zF(w) = \frac{F(x + yw) - F(z)}{yF'(z)}, \quad z \cdot w = x + yw, \quad A_z\Phi(F) = |Q_F(z)|^2\Phi(T_zF). \tag{4}$$

Direct differentiation gives

$$Q_{T_zF}(w) = Q_F(z \cdot w)/Q_F(z), \qquad T_w(T_zF) = T_{z\cdot w}F, \qquad A_zA_w = A_{z\cdot w}.$$

On $C(\mathcal{K})$ with the supremum norm, the positive bounded operator is

$$L = \tfrac{1}{2}(A_{-1/2+i/2} + A_{1/2+i/2}), \quad L^n = v\sum_{j=0}^{2^n-1} A_{-1+(2j+1)v+iv}, \quad v = 2^{-n}. \tag{5}$$

Weights telescope, so the dyadic identity follows inductively. Positivity gives $\|L^n\| = \max_F L^n 1(F)$, and submultiplicativity gives $\rho = \lim_n \|L^n\|^{1/n} = \inf_n \|L^n\|^{1/n}$. The source's disk comparison uses the exact Cayley factor

$$|f'(m(z))|^{-2} = 16|Q_F(z)|^2/|z + i|^4.$$

Fixed compact-set distortion compares arcs at height comparable to $1 - r$ to the dyadic samples; four rotations cover the circle. Therefore $\rho \le 2$ implies (2)'s epsilon-loss estimate. No positive eigenfunction, compact transfer operator, or spectral gap is presupposed here.

2.2 A law under the contrary growth hypothesis

Suppose $\rho > 2$, and write $\beta = \log_2 \rho > 1$. The source constructs a probability law P on $\mathcal{K}$ with

$$\mathbb{E}_P\big[|Q_F(x + iy)|^2\Phi(T_{x+iy}F)\big] = y^{-\beta}\mathbb{E}_P\Phi(F) \tag{6}$$

for each fixed $x + iy$ and every nonnegative Borel Φ. This is weighted covariance, not invariance of P under unweighted rerooting.

The construction has three indispensable steps. For $\tau > \rho$, the positive resolvent $R_\tau = \sum_{n\ge0} \tau^{-n}L^n$ exists. Put $g_\tau = R_\tau 1$, $M_\tau = \max g_\tau$. The inequality $Lg_\tau = \tau(g_\tau - 1) \le \tau(1 - M_\tau^{-1})g_\tau$ shows $M_\tau \to \infty$ as $\tau \downarrow \rho$: a bounded subsequence would contradict the defining spectral growth after taking roots. Normalized maximizing resolvent evaluations converge weakly to a positive eigenmeasure $\mu L = \rho\mu$.

Right composition of (5) by A_{2+i} cancels all but two grid endpoints. Uniform interior distortion bounds the remaining difference by $C(2/\rho)^n$. Thus $\mu A_{2+i} = \mu$ under the strict contrary hypothesis. Averaging over horizontal period two and then one logarithmic scale period gives (6). Equality of finite

Borel measures extends the continuous-test identity to nonnegative Borel tests. At $\rho = 2$, the endpoint cancellation need not vanish; the construction is not asserted at that endpoint and is not needed there.

Define $q_F = iyQ'_F/Q_F = a_F + ib_F$. Since the normalized reciprocal derivatives and their derivatives are uniformly bounded on compact subsets, differentiation under expectation is legitimate. From $\mathbb{E}|Q_F(x+iy)|^2 = y^{-\beta}$ and $Q_F(i) = 1$ one obtains

$$\mathbb{E}a(i) = -\beta/2, \quad \mathbb{E}b(i) = 0, \quad \mathbb{E}|q(i)|^2 = \beta(\beta+1)/4. \tag{7}$$

Indeed $\partial_y|Q|^2 = 2a|Q|^2/y$, $\partial_x|Q|^2 = 2b|Q|^2/y$, and $\Delta|Q|^2 = 4|Q'|^2$.

2.3 Escape, topology, and transport

Set $k = (\beta+3)/4 > 1$ and

$$V_\xi^F(z) = y^k/|F(z) - \xi|, \qquad G_F(z) = F(z) - iyF'(z)/k, \qquad J_F = \det DG_F/|F'|^2.$$

For almost every sampled map and almost every target, the source proves compactness of each closed positive superlevel of V. A crucial weighted inverse-area estimate uses the affine inverse $(x,y) \mapsto (-x/y, 1/y)$, whose absolute Jacobian is y^{-3}. A horizontal trace estimate then controls small-height escape. These are global analytic facts; finite coefficient panels cannot replace their tail conditions.

At a critical point with $G_F(z) = \xi$, direct real differentiation gives

$$k^2 J = k(k-1) + (2k-1)a + |q|^2, \tag{8}$$

$$\operatorname{Hess}\log V = \frac{k}{y^2}\begin{pmatrix} -(k+a) & b \\ b & k-1+a \end{pmatrix}, \qquad \det \operatorname{Hess}\log V = -k^4 J/y^4. \tag{9}$$

The negative trace implies that $J < 0$ corresponds to a maximum and $J > 0$ to a saddle. Fix a good target: its positive superlevels are compact and ξ is a regular value of G_F, so the critical points are nondegenerate. Away from the possible pole, $\Delta \log V = -k/y^2 < 0$ rules out an interior minimum enclosed by a level curve. Thus a regular superlevel component has no holes; the pole, if present, lies in the superlevel. The component count then satisfies $C = N_{\max} - N_{\text{saddle}} + B$, where $B \in \{0,1\}$ is the possible pole. Since $C \geq B$, there are at least as many maxima as saddles above each level. The pole is not counted as a maximum. Ordering by decreasing V and then lexicographic source coordinates pairs saddles injectively to maxima of at least the same value. Countable inverse charts establish measurability, and the order respects positive source dilations and horizontal translations.

The transport weight is $\omega = y^{\beta-1}|Q|^2$. Changing variables through the sign charts of G_F produces the target weight

$$y^{\beta-1}|Q|^4 = (V/k)^4 \tag{10}$$

at a critical point, since $4(k-1) = \beta - 1$. The fourth power is essential: the second $|Q|^2$ comes from dividing by $|\det DG|$. The pairing increases this target weight. Although good targets have full area measure, the root target $G_F(i) = -i/k$ is fixed. A separate source-box/area-formula argument establishes that its exceptional positive-J weight is zero. Ignoring this step would leave a genuine logical omission; M150 includes it.

For paired points restrict displacement by $N^{-1} \leq y'/y \leq N$, $|x'-x| \leq Ny$, and denote the selected set by D_N. The source box $|x| \leq R$, $1 \leq y \leq Y$ maps into $|x| \leq R + NY$, $1/N \leq y \leq NY$. Covariance yields expected density $\mathrm{d}x\,\mathrm{d}y/y$, not hyperbolic area. Consequently

$$\mathbb{E}[J_+(i)1_{D_N}] \leq c_{N,R,Y}\mathbb{E}J_-(i), \qquad c_{N,R,Y} = \left(1 + \frac{NY}{R}\right)\left(1 + \frac{2\log N}{\log Y}\right). \tag{11}$$

Hold N fixed, choose $R = Y^2$, let $Y \to \infty$, then let $N \to \infty$. Monotone convergence and the root-exception lemma give $\mathbb{E}J \le 0$. But (7) and (8) give

$$\mathbb{E}J = (k-1)/k = (\beta-1)/(\beta+3) > 0. \tag{12}$$

This reconstructs the source's contradiction with its measure factors and limit order visible. It is an explanation of the audited chain, not a substitute for its escape and measurable-selection proofs.

2.4 Area range and endpoint limits

For $-2 < t < 0$, put $q = -t/2 \in (0,1)$. Holder gives $I_f(t,r) \le I_f(-2,r)^q$. Choosing $q(1+\epsilon) < 1$ in the source bound makes the radial area integral finite. For $0 < t < 2/3$, the source uses the classical estimate $\int_{\mathbb{D}}(1+|f|)^a \,\mathrm{d}A < \infty$ for $a < 1$. Choose $\eta > 0$ so that $a = (2+\eta)t/(2-t) < 1$. Injectivity gives

$$\int_{\mathbb{D}} |f'|^2(1+|f|)^{-2-\eta}\,\mathrm{d}A \le \int_{\mathbb{C}}(1+|w|)^{-2-\eta}\,\mathrm{d}A = \frac{2\pi}{\eta(1+\eta)}.$$

Holder with exponents $2/t$ and $2/(2-t)$ then bounds $\int |f'|^t$. This positive-power argument is separate from the new negative endpoint. The Koebe map has logarithmic area divergence at both $t = -2$ and $t = 2/3$; it prevents extending the universal open range by adding endpoints.

3 Exact Ram Log-Partition Coordinates

Since f' is nonzero on the simply connected disk, choose analytic $g = \log f'$ with $g(0) = 0$. At fixed $r < 1$ set

$$X(\theta) = \operatorname{Re} g(re^{i\theta}), \qquad \Lambda(p) = \log \mathbb{E}_\theta e^{pX}, \qquad \mathrm{d}\nu_s = e^{sX - \Lambda(s)}\frac{\mathrm{d}\theta}{2\pi}.$$

The harmonic mean-value property gives $\mathbb{E}_\theta X = 0$. Boundedness of X on the fixed interior circle permits all real differentiations:

$$\Lambda'(s) = \mathbb{E}_{\nu_s} X, \quad \Lambda''(s) = \operatorname{Var}_{\nu_s} X, \quad \log I_f(-2,r) = \int_{-2}^{0} (s+2)\operatorname{Var}_{\nu_s} X \,\mathrm{d}s. \tag{13}$$

The last equality is Taylor's integral formula using $\Lambda(0) = \Lambda'(0) = 0$. It requires no convergent complex cumulant series and no Gaussian assumption. It holds for the full class $\mathcal{S}$. The tilt ν_s is a probability on angles for one map and radius; it must not be identified with the source's contrary-growth law P on maps.

Proposition 3.1 (Exact averaged endpoint criterion)**.** *The upper bound $B_{\mathcal{S}}(-2) \le 1$ is equivalent to requiring, for every fixed $f \in \mathcal{S}$,*

$$\limsup_{r\uparrow 1} \frac{1}{T}\int_{-2}^{0} (s+2)\operatorname{Var}_{\nu_s} X \,\mathrm{d}s \le 1.$$

A universal epsilon-loss constant instead follows from the corresponding uniform finite-scale bound, including a uniform additive constant.

Proof. Substitute (13) into the definition (1) and take the supremum only after each fixed-map limsup. For the uniform version, an estimate of the integral by $(1+\epsilon)T + C'_\epsilon$ exponentiates to (2) with $C_\epsilon = e^{C'_\epsilon}$. □

This is an equivalent representation of the difficult estimate, not an independent proof of it. It is weaker and more realistic than bounding every tilted variance separately. The distinction is forced by the next example, not merely a preference of formulation.

4 A Critical-Tilt Obstruction to an Earlier Premise

Theorem 4.1 (Quadratic calibration and concentrated curvature)**.** *The bounded map $f_2(z) = z - z^2/2$ lies in $\mathcal{S}_b$ and satisfies*

$$I_{f_2}(-2, r) = \frac{1}{1-r^2}, \qquad \mathrm{Var}_{\nu_{-1}} \log |f_2'(re^{i\theta})| = \frac{T^2}{12} + O(T). \tag{14}$$

In particular, the premise $\sup_{|s|\le p_0} \mathrm{Var}_{\nu_s} X/T \le 1/2 + o(1)$ from TR-item-0ΥΥ-astra, equation (16), cannot hold for every bounded univalent map when $p_0 \ge 1$.

Proof. Equality of two values of f_2 implies $(z-w)(1-(z+w)/2) = 0$. Since $|z+w| < 2$ in $\mathbb{D}$, it implies $z = w$. Normalization and boundedness are immediate. Parseval applied to $(1-z)^{-1}$ gives the first identity in (14).

Put $\delta = 1 - r$, $d(\theta) = |1 - re^{i\theta}|$ and

$$A_m = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{(\log d(\theta))^m}{d(\theta)} \, \mathrm{d}\theta, \qquad m = 0, 1, 2.$$

Fix a small positive c. On $[\delta, c]$, $d^2 = \delta^2 + 4r \sin^2(\theta/2)$ and comparability with θ imply

$$|\log(d/\theta)| + |\theta/d - 1| \le C(\delta + (\delta/\theta)^2 + \theta^2).$$

The bound is uniform for sufficiently small δ. Replacing d by θ in the integrand produces an integrated error $O(T^m)$: the term $(\delta/\theta)^2$ is integrable with respect to $\mathrm{d}\theta/\theta$ and contributes at most CT^m; the θ^2 term has bounded logarithmic moments; and $\delta T^{m+1} = O(1)$. Polynomial expansion handles the logarithm numerator, including $m = 0$. On $[0, \delta]$, d is comparable to δ, so the contribution is $O(T^m)$. On $[c, \pi]$ all terms are uniformly bounded. Evenness and $u = \log(1/\theta)$ therefore give

$$A_0 = T/\pi + O(1), \quad A_1 = -T^2/(2\pi) + O(T), \quad A_2 = T^3/(3\pi) + O(T^2). \tag{15}$$

Under ν_{-1} the mean is $A_1/A_0 = -T/2 + O(1)$ and the second moment is $A_2/A_0 = T^2/3 + O(T)$. Subtracting the square proves the variance formula. Dividing by T disproves the specified premise. □

This does not contradict Brennan, the source manuscript, or the conditional logic of the earlier theorem. It demonstrates that the proposed premise is unavailable on its intended full class at these tilts. The exact identity in that report remains correct, and the present report records the correction without altering the original private document.

Proposition 4.2 (An exact critical-tilt symmetry)**.** *For the same quadratic map and every $0 < r < 1$,*

$$\mathbb{E}_{\nu_{-1}} X = \tfrac{1}{2} \log(1 - r^2), \qquad I_{f_2}(p, r) = (1 - r^2)^{p+1} I_{f_2}(-p-2, r).$$

Proof. On the unit circle use the involution $w = (r - z)/(1 - rz)$ and put $a = 1 - r^2$. Then $|1 - rw| = a/|1 - rz|$ and $|\,\mathrm{d}w| = a|\,\mathrm{d}z|/|1 - rz|^2$. Thus $|\,\mathrm{d}w|/|1 - rw| = |\,\mathrm{d}z|/|1 - rz|$: the critical tilted measure is invariant, while X changes to $\log a - X$. Taking its mean proves the first formula; substituting in the unweighted power integral proves the second. This exact symmetry also checks the asymptotic normalization. □

For the same map $\log I(-2, r) = T - \log(1 + r)$, so the averaged quantity in Proposition 3.1 tends to one despite the large variance at $s = -1$. More generally elementary integration gives

$$\beta_{f_2}(p) = \max\{-p - 1, 0\}.$$

For $p > -1$ the mean stays bounded; at $p = -1$ it grows logarithmically; for $p < -1$ it is comparable to δ^{p+1}. The corner at $p = -1$ explains concentrated curvature in the limiting log-partition. Meanwhile the untilted variance is only

$$\operatorname{Var}_\theta X = \frac{1}{2}\sum_{n\geq 1}\frac{r^{2n}}{n^2} \leq \frac{\pi^2}{12}.$$

Thus ordinary variance or a suppression of all higher cumulants cannot recover this endpoint behavior by itself. A successful Ram representation must retain rare negative-derivative events and cross-scale effects.

5 Transfer Certificates and a Moment-Matching Obstruction

5.1 Positive certificates in the correct norm

Theorem 5.1 (Superfunctions with residuals)**.** *Let L be a positive bounded operator on $C(\mathcal{K})$ and $h \in C(\mathcal{K})$ with $0 < m \leq h \leq M$. If $Lh \leq \lambda h + \eta$ pointwise, with $\eta \geq 0$ and $\lambda + \eta/m > 0$, then*

$$\|L^n\| \leq (M/m)(\lambda + \eta/m)^n, \qquad \rho \leq \lambda + \eta/m.$$

If $\lambda > 0$ and $\left\|L^N\right\| \leq \lambda^N$, then $h = \sum_{j=0}^{N-1} \lambda^{-j} L^j 1$ satisfies $Lh \leq \lambda h$.

Proof. Since $\eta \leq (\eta/m)h$, positivity and $1 \leq h/m$ imply $L^n 1 \leq m^{-1}(\lambda + \eta/m)^n h$. Take the supremum and roots. For the second assertion, summation telescopes to $Lh = \lambda(h - 1 + \lambda^{-N} L^N 1) \leq \lambda h$. □

The premise must hold on all of $\mathcal{K}$, not only sampled conformal maps. Computed map samples usually give lower tests for $\left\|L^N\right\|$. One fixed $\lambda > 2$ is nonsharp; certificates for every $\lambda > 2$ would give $\rho \leq 2$. Coordinate changes must retain the supremum norm or prove the necessary norm comparison. A finite L^2 matrix is not automatically this operator.

Proposition 5.2 (Exact lower spectral calibration)**.** *For the operator* (5)*, $\rho \geq 2$ without using the source's contradiction argument.*

Proof. Take $f_+(z) = z + z^2/2$, whose univalence follows as for f_2, and $F(z) = 2if_+(m(z))$. Differentiation gives $Q_F(z) = -(z+i)^3/(8z)$. At the two dyadic samples $z = \pm v + iv$, $|z + i| \geq 1$ and $|z|^2 = 2v^2$, so $|Q_F(z)|^2 \geq 1/(128v^2)$. Both samples occur for $n \geq 1$. Hence $L^n 1(F) \geq 1/(64v) = 2^n/64$, and taking roots proves the assertion. □

Thus no sound certificate can give $\rho < 2$. If the source's upper proof is accepted, this supplies the exact calibration $\rho = 2$ for its operator.

5.2 Why root moments cannot replace covariance

Proposition 5.3 (A feasible finite law with the wrong transformation law)**.** *There is a probability on genuine normalized univalent maps with the three root moments* (7) *at $\beta = 3/2$, but without* (6)*.*

Proof. If $f(z) = z + a_2 z^2 + \cdots$, the corresponding half-plane map has $q_F(i) = 1 - a_2$. Give the Koebe map $z/(1-z)^2$ mass $6/7$ and $z/(1 - z/4)$ mass $1/7$. Their $q(i)$ values are -1 and $3/4$, respectively; therefore

$$\mathbb{E}a = -3/4, \quad \mathbb{E}b = 0, \quad \mathbb{E}|q|^2 = 15/16.$$

These are precisely the required moments at $\beta = 3/2$. At $z = 2i$, the Cayley coordinate is $1/3$ and $Q_F(2i) = 9/(4f'(1/3))$. The respective values are $1/2$ and $121/64$. Consequently

$$\mathbb{E}|Q_F(2i)|^2 = \frac{6}{7}\frac{1}{4} + \frac{1}{7}\frac{14641}{4096} = \frac{20785}{28672} \neq 2^{-3/2}.$$

The constant test in (6) already fails. □

This is not a counterexample to the source's law, which is constructed under its contrary hypothesis with far more information. It is a direct test against a proposed replacement by finitely many fitted moments. Even exact positive physical moment matrices cannot recover an absent continuum covariance identity.

6 Certified Ram Panels

6.1 Reciprocal-derivative coefficients

Write $H = 1/f' = \sum_{n\geq 0} c_n z^n$. At every $r < 1$, Parseval gives

$$I_f(-2, r) = \sum_{n\geq 0} |c_n|^2 r^{2n}. \tag{16}$$

For this endpoint, direct reciprocal-derivative panels are simpler than reconstructing all tilted variances. If $r < R < 1$, distortion gives $\max_{|z|=R} |H(z)| \leq B_R = (1 + R)^3/(1 - R)$. Parseval also at R yields

$$\sum_{n>N} |c_n|^2 r^{2n} \leq (r/R)^{2N+2} \left(B_R^2 - \sum_{n=0}^{N} |c_n|^2 R^{2n} \right) \leq B_R^2 (r/R)^{2N+2}. \tag{17}$$

Indeed each omitted $(r/R)^{2n}$ is at most $(r/R)^{2N+2}$, and $\sum_{n\geq 0} |c_n|^2 R^{2n} \leq B_R^2$. The bracketed refinement uses exact retained coefficients (or a certified lower bound for their energy). A coefficient-by-coefficient Cauchy estimate followed by a geometric sum would give the weaker last bound divided by $1 - (r/R)^2$. The physical energy budget avoids that loss under the same assumptions. No unknown boundary regularity is assumed; all contours are interior. If a computed polynomial $\widehat{H}_N$ has a certified total error $\left\| H - \widehat{H}_N \right\|_{L^2(r)} \leq e_r$, triangle inequalities give

$$(\max\{0, \left\| \widehat{H}_N \right\|_{L^2(r)} - e_r\})^2 \leq I_f(-2, r) \leq (\left\| \widehat{H}_N \right\|_{L^2(r)} + e_r)^2. \tag{18}$$

Coefficient errors, truncation, and quadrature must all be included in e_r in the same normalized angular norm. Equations (17)– (18) are finite-radius guarantees for an admissible map; they do not cover the universal class merely because B_R is uniform. The bound worsens as $r, R \to 1$, and finite arbitrary coefficient vectors need not arise from the reciprocal derivative of a globally univalent map.

For an invertible finite synthesis $c = Ru$ in Ram coordinates, the radial mass is $R^* \operatorname{diag}(1, r^2, \ldots, r^{2N})R$, not identity. TR248's finite-frame error budgets and TR250's transformed adjoints retain this physical norm. They do not turn a finite reconstruction theorem into an exact frame for every analytic boundary field.

6.2 Log-field panels and asymptotic budgets

If $g = \log f'$ and $\|g - \widehat{g}\|_{L^\infty(r)} \leq \eta$, then pointwise exponential comparison proves

$$e^{-2\eta} I_{\widehat{g}}(-2, r) \leq I_f(-2, r) \leq e^{2\eta} I_{\widehat{g}}(-2, r), \qquad |\Lambda(-2) - \widehat{\Lambda}(-2)| \leq 2\eta.$$

To transfer an exponent without asymptotic loss, $\eta/T \to 0$ suffices together with a genuine bound on the reconstructed means. A nonvanishing error can instead be charged to an allowed exponent budget; the stated no-loss condition is not a necessary condition for every possible estimate. An L^2 error for g alone does not control the exponential of its negative part: rare angular sets can dominate. Unweighted least squares on g therefore does not certify the endpoint. For an explicit demonstration, fix $r_0 \in (0,1)$, take the actual map $f(z) = z$ with $g = 0$, and let $g_N(z) = -N^{-3/4}\sum_{j=1}^{N}(z/r_0)^j$. On $|z| = r_0$, $\|g_N\|_{L^2} = N^{-1/4} \to 0$, but for $|\theta| \le 1/(3N)$, $-2\operatorname{Re} g_N(r_0 e^{i\theta}) \ge N^{1/4}$. Consequently the reconstructed inverse-square mean is at least $e^{N^{1/4}}/(3\pi N) \to \infty$, whereas the true mean is one. These polynomial approximants are not asserted to be logarithmic derivatives of univalent approximating maps; they show precisely why the proposed panel error datum is insufficient.

6.3 A separate Schwarzian-multiplier route

The normalized weighted Bergman space A^2_α, $\alpha > -1$, has monomial masses

$$m_n(\alpha) = (\alpha+1)\int_0^1 t^n(1-t)^\alpha \,\mathrm{d}t = \frac{n!\,\Gamma(\alpha+2)}{\Gamma(n+\alpha+2)}.$$

For the actual Schwarzian $S(f) = f'''/f' - \frac{3}{2}(f''/f')^2$, multiplication maps A^2_α to $A^2_{\alpha+4}$. Shimorin's sufficient criterion, recalled in Jin's Proposition 1.4(2) [3], uses, for every $\alpha > 0$,

$$\left\|M_{S(f)}\right\|^2 \le \frac{36(\alpha+3)(\alpha+5)}{(\alpha+2)(\alpha+4)} \quad\Longrightarrow\quad \beta_f(-2) \le 1.$$

The implication is an external theorem, not proved here. A finite polynomial domain with coefficient multiplication matrix A must use the generalized Rayleigh quotient $v^*A^*M_{\alpha+4}Av/(v^*M_\alpha v)$. Restricting domain modes gives a lower test of the full norm; truncating output also discards positive mass. Neither is the required upper bound without certified remaining input and output blocks. Stable Ram coordinates improve the bookkeeping and may facilitate certified computation, but do not provide that missing continuum multiplier inequality.

7 A Quantitative Transport Extension

The source proves its contradiction by ordered exact limits. A useful Ram residual formulation exposes the margins needed for a finite-box certificate instead of silently discarding errors.

Theorem 7.1 (Finite-box obstruction with explicit defects)**.** *Fix $\beta > 1$, $k = (\beta+3)/4$, $N \ge 1$, $R > 0$, $Y > 1$. Suppose integrable real variables a, b define $q = a + ib$ and J by* (8)*, and suppose*

$$|\mathbb{E}a + \beta/2| \le e_a, \qquad |\mathbb{E}|q|^2 - \beta(\beta+1)/4| \le e_2.$$

Let D_N be a measurable selection for which

$$\mathbb{E}[J_+ 1_{D_N}] \le c_{N,R,Y}\mathbb{E}J_- + e_{\mathrm{tr}}, \qquad \mathbb{E}[J_+ 1_{D_N^c}] \le \tau_N, \qquad e_{\mathrm{tr}}, \tau_N \ge 0.$$

Then necessarily

$$\frac{\beta-1}{\beta+3} \le \frac{(2k-1)e_a + e_2}{k^2} + (c_{N,R,Y}-1)\mathbb{E}J_- + e_{\mathrm{tr}} + \tau_N. \tag{19}$$

Proof. Taking expectations in (8), its exact baseline is $(k-1)/k$. The stated moment error bounds therefore imply $\mathbb{E}J \ge (\beta-1)/(\beta+3) - ((2k-1)e_a + e_2)/k^2$. On the other hand, split $\mathbb{E}J_+$ over D_N and its complement and use the transport and tail inequalities. Subtracting $\mathbb{E}J_-$ gives $\mathbb{E}J \le (c-1)\mathbb{E}J_- + e_{\mathrm{tr}} + \tau_N$. Combine them. □

This theorem applies to any objects satisfying its explicit premises; it does not assert that finite samples certify them. In an implementation, e_{tr} must enclose normalized area change, covariance, selection, and integration defects, while τ_N includes both displacement tails and bad-root mass. A rigorous upper bound on $\mathbb{E}J_-$ is also necessary. Interval arithmetic on a few jets is insufficient for these obligations.

For a desired box excess $\zeta > 0$, one can impose $NY/R \le \zeta/3$ and $2\log N/\log Y \le \zeta/3$; if $\zeta \le 3$, then $c - 1 \le \zeta$. This gives explicit, potentially very large required boxes. Near $\beta = 1 + \delta$, the available contradiction margin is $\delta/(4 + \delta)$. A fixed numerical tolerance cannot exclude every $\delta > 0$. The residual theorem clarifies how adaptivity must scale; it does not claim a computational speedup or eliminate the source's exact all-scale method.

8 Further Deductions and Limits

8.1 Loss-free endpoint bounds require another theorem

Bertilsson's Theorem 3.7(a),(f) [8] equates the endpoint spectrum condition with a uniform loss-free integral-means estimate. Conditionally on that external equivalence and M150's $B_{\mathcal{S}}(-2) = 1$, it yields

$$I_f(-2, r) \le C(1 - r)^{-1} \quad (f \in \mathcal{S}).$$

This is stronger than the explicitly epsilon-loss bound, but it is not obtained by setting $\epsilon = 0$ in C_ϵ. The theorem statement and its hypotheses were checked in the primary thesis; its complete dandelion argument was not independently audited here. This strengthened corollary is accordingly recorded with that external dependency rather than presented as a new self-contained proof.

8.2 Negative tails and bounded versus full classes

Distortion gives $|f'|^{-1} \le 8/(1 - r)$ on $r \ge 1/2$. Combining with (2), for $p \le -2$,

$$I_f(p, r) \le 8^{|p|-2} C_\epsilon (1 - r)^{-(|p|-1)-\epsilon}.$$

The quadratic f_2 supplies the matching exponent $|p| - 1$. This reproduces the source's tail consequence with a bounded explicit lower witness. It does not establish a quadratic central branch. For positive exponents the class distinction is substantial: the Koebe map has $\beta(2) = 5$ in the full class, whereas the known bounded-class supremum at 2 is 1. TR-item-077's positive branch cannot be transferred by changing its label to $\mathcal{S}$. Likewise its fixed bounded-map upper support bound for X/T is $1 + o(1)$, while full-class distortion permits $3 + o(1)$. These distinctions remain necessary regardless of the status of the new Brennan claim.

9 Comparison, Research Directions, and Conclusion

9.1 Advantages and costs

The source's strength is geometric: it constructs the weighted law and uses compact superlevels, topology, measurable pairing, and exact ordered limits to force a sharp spectral contradiction. The Ram representations make different parts inspectable. Log-partition differentiation isolates tilted fluctuations without unjustified cumulant convergence. Reciprocal coefficients turn the actual endpoint integral into a positive physical quadratic form. Positive superfunctions give a clear upper-certificate target and residual tolerance. The finite-box theorem exposes how moment, transport, and tail errors must fit within one common contradiction margin.

The disadvantages are equally concrete. Sharp all-map bounds do not follow from finite moment fitting; the explicit two-map law demonstrates this. Critical-tilt variance can grow quadratically in scale, invalidating a naive Gaussian-style premise. Boundary tails make truncation expensive. No measured runtime improvement, sharp universal superfunction, or finite certificate for all univalent maps has been produced here. Classical complex analysis and the source's topology cannot be removed by algebraic coordinate equivalence alone.

9.2 Focused future work

1. Develop adaptive tilt quadrature that preserves concentrated curvature, with rigorous errors for the weighted integral in (13). The quadratic example must remain a mandatory analytic calibration.
2. Construct candidate positive superfunctions on certified covering sets of univalent maps and bound the omitted analytic tails. Report subclass certificates separately from universal ones.
3. Obtain quantitative escape and displacement-tail bounds in geometrically controlled subclasses; combine them with Theorem 7.1.
4. Extend compatible Ram panels to the full Schwarzian multiplication operator, with rigorous input-block and output-tail norm bounds rather than finite-section eigenvalues alone.
5. Independently audit the loss-free equivalence before incorporating it as an unconditional improvement of the source's stated radial estimate.

9.3 Conclusion

M150 is an L5 match because the Ram log-partition field is exactly the source's integral-means observable, not merely a related analogy. The comparison yields a proved correction to an earlier proposed closure premise, exact spectral and moment calibrations, and useful certified finite-scale and defect-budget extensions. It also pinpoints what remains nontrivial: universal univalence constraints, full affine covariance, escape and selection, and vanishing all-scale errors. These results support further transparent analysis without relabeling a conditional criterion as a complete new solution to Brennan's conjecture.

A Reproducibility and Source Locators

The public source directory is `Brennans-conjecture-and-sharp-inverse-square-integral-means-September-24-2026`. The central labels are `lem:transfer-growth`, `prop:disk-transfer`, `lem:weighted-law`, `lem:law-moments`, `lem:weighted-inverse-area`, `prop:proper-superlevels`, `lem:critical-count`, `prop:critical-pairing`, `lem:root-exception`, and `prop:transfer-bound`. The full task-local proof audit is `../_audit/M150-proof-audit.txt`; its public-literature section records the depth of each external check. No formalization or its axiom coverage has been certified. The local computations are proved in the text; finite arithmetic checks are regression tests, not replacements. The exact-rational script `../_audit/check_M150.py` checks the moment cancellation, two-map law, dyadic lower tests, and box-factor algebra; its saved results passed. The independent note `../_audit/M150-local-audit.txt` reproduces the critical-tilt error orders, exact symmetry, stronger energy tail, and the other local implications, with their limitations.

The specific private inputs are:

- `C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/res-math/TR-item-077-astra.tex`, especially equations (1)–(7), (15)–(20), and the fixed-function quantifier counterexample. Its equation (16) is corrected in scope here.

- C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/TechReps/FinalTechReps/TRs/TR248-Stable-Adaptive-Ram-Frames/TR248-Stable-Adaptive-Ram-Frames.tex, finite reconstruction, Parseval, and perturbation statements.
- C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/TechReps/FinalTechReps/TRs/TR250-Compatible-Variational-Ram-Panels/TR250-Compatible-Variational-Ram-Panels.tex, finite transfer and physical mass matrices.

Only public titles, identifiers, and URLs were used in public literature queries. Private corpus text was not uploaded as research material.

References

[1] OpenAI. *Brennan's conjecture and sharp inverse-square integral means*. September 24, 2026; released October 6, 2026. Pinned snapshot specified in the text. https://github.com/openai/math.

[2] J. Jin. *Complex exponential integral means spectra of univalent functions and the Brennan conjecture*. arXiv:2512.09330v13, June 5, 2026. https://arxiv.org/html/2512.09330v13.

[3] J. Jin. *On the norms of the multiplication operators between weighted Bergman spaces*. arXiv:2603.16170v9, May 16, 2026. https://arxiv.org/html/2603.16170v9.

[4] A. Aleman and A. Kouroupis. *Brennan's conjecture holds for semigroups of holomorphic functions*. arXiv:2409.15074v2, April 14, 2025. https://arxiv.org/html/2409.15074v2.

[5] Y. Zheng. *Brennan Conjecture for Basin of Attraction at Infinity*. arXiv:2604.12240v1, April 14, 2026. https://arxiv.org/html/2604.12240v1.

[6] H. Hedenmalm and S. Shimorin. *Weighted Bergman spaces and the integral means spectrum of conformal mappings*. Duke Mathematical Journal 127 (2005), 341–393. https://arxiv.org/abs/math/0406345.

[7] A. Sola. *Bergman space methods and integral means spectra of univalent functions*. KTH licentiate thesis, 2007, pp.5–6. https://www.diva-portal.org/smash/get/diva2:11935/FULLTEXT01.pdf.

[8] D. Bertilsson. *On Brennan's conjecture in conformal mapping*. KTH doctoral thesis, 1999, Theorem 3.7, pp.44–45. https://www.diva-portal.org/smash/get/diva2:8593/FULLTEXT01.pdf.

[9] M. Subbarao. *Ram Calculus for the Universal Integral-Means Spectrum: Corrected Formulation, Multiscale Cumulants, and the BCJK–Brennan Gap*. TR-item-077-astra, September 6, 2026. Confidential local corpus.

[10] M. Subbarao. *Stable Adaptive Ram Frames*. TR248. Confidential local corpus; finite reconstruction passages listed above.

[11] M. Subbarao. *Compatible Variational Ram Panels*. TR250. Confidential local corpus; physical norm and transfer passages.